Program -1

Identification and Retrieval of Datasets

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Aim

Write a Python program to demonstrate how to identify and retrieve datasets from Kaggle and perform basic exploration using Python libraries such as Pandas and NumPy.

Algorithm

1. Start.
2. Import Pandas and NumPy.
3. Load Iris.csv into a DataFrame.
4. Display the first five and last five records.
5. Display the number of rows and columns.
6. Display the column names and data types.
7. Display dataset information.
8. Check and display missing values.
9. Check and display duplicate records.
10. Generate summary statistics.
11. Select numerical columns.
12. Calculate the mean and standard deviation using NumPy.
13. Find the number of unique values in each column.
14. Display complete dataset statistics.
15. Stop.

Program - 1

```
import pandas as pd
import numpy as np

df = pd.read_csv("Iris.csv")

print("Dataset loaded successfully!")

print("\nFirst Five Records:")
print(df.head())

print("\nLast Five Records:")
print(df.tail())

print("\nDataset Shape:")
print(df.shape)

print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])

print("\nFeature Names:")
print(df.columns.tolist())


print("\nData Types:")
print(df.dtypes)

print("\nDataset Information:")
df.info()

print("\nMissing Values:")
print(df.isnull().sum())

print("\nNumber of Duplicate Records:")
print(df.duplicated().sum())

print("\nSummary Statistics:")
print(df.describe())



# Select numerical columns
numeric_data = df.select_dtypes(include=np.number)

print("\nNumerical Columns:")
print(numeric_data.columns.tolist())
```

```
# Calculate mean using NumPy
print("\nMean of Numerical Features:")
print(np.mean(numeric_data, axis=0))

# Calculate standard deviation using NumPy
print("\nStandard Deviation:")
print(np.std(numeric_data, axis=0))

print("\nUnique Values in Each Column:")

for column in df.columns:
    print(column, ":", df[column].nunique())

print("\nDataset Description:")
print(df.describe(include='all'))
```